River and lake condition: Pune and Pimpri-Chinchwad

Jal-drishti AI · Sentinel-2 · latest month 2026-10

Item	Value
Water bodies	Mula, Mutha, Mula-Mutha and Pavana rivers; Pashan and Katraj lakes
Stretches	158 (1,139 ha of water)
Sample first / watch	13 / 41
Floating weed now	8.0% of visible water (90.4 ha)
Months analysed	2024-11 to 2026-10

Method

Channel: Open water (MNDWI > 0) in any Sep-Nov Sentinel-2 composite 2019-2025; water bodies under 2 ha dropped.

Stretches: Channel split on a 0.01° (~1.1 km) grid; stretches under 1 ha dropped; named after the nearest Census village.

Hyacinth: Share of the cloud-free channel with NDVI above 0.3 (floating vegetation), excluding ESA WorldCover trees and mangroves.

Turbidity chlorophyll: NDTI and NDCI over open water away from the banks (one 20 m pixel in from the edge, to avoid mixed land-water pixels), compared with the same month in 2019-2023 at the same pixels (z-score).

Sample first: Hyacinth over 50% of the channel, or turbidity or chlorophyll 2+ standard deviations above normal in the latest clear month.

Limits: Optical proxies only: BOD, dissolved oxygen and coliforms need samples, and index values need local calibration with them (algae types differ by water body). Sentinel-2 L2A uses a land-oriented atmospheric correction, which is less accurate over dark water (Palacios et al. 2015); a water-specific correction such as ACOLITE is a planned upgrade. Channels narrower than ~40 m have no interior pixels for water colour. Tides affect creeks.

Results

Stretch	Status	Why	Month	Water area (ha)	Hyacinth %	Turbidity (NDTI)	Turbidity vs normal (z)	Chlorophyll (NDCI)	Chlorophyll vs norma
S062 · near Pimpale Gurav	sample first	heavy hyacinth	2026-10	2.78	97.1				
S116 · near Fursungi	sample first	heavy hyacinth	2026-10	1.77	81.8				
S111 · near Wagholi	sample first	heavy hyacinth	2026-10	1.89	59.6				
S078 · city	sample first	unusually green water	2026-10	1.53	18.4	-0.113	-0.89	0.1301	3.17
S047 · city	sample first	unusually green water	2026-10	2.82	12.9	-0.1088	-0.65	0.1254	2.96
S012 · Lavale	sample first	unusually green water	2026-10	2.61	9.2	-0.2515	-2.8	0.3246	6.67

Stretch	Status	Why	Month	Water area (ha)	Hyacinth %	Turbidity (NDTI)	Turbidity vs normal (z)	Chlorophyll (NDCI)	Chlorophyll vs norma
S121 · Pimpalgaon Tarf Chakan	sample first	unusually green water	2026-10	8.25	8.3	-0.1102	-1.07	0.1716	2.61
S034 · near Bhugaon	sample first	unusually green water	2026-10	2.66	7.6	-0.1682	-2.54	0.0998	2.61
S158 · Kadamwak Wasti	sample first	unusually green water	2026-10	2.49	4.8	-0.2636	-2.42	0.3376	6.45
S151 · Kolwadi	sample first	unusually green water	2026-10	25.79	4.4	-0.2382	-2.58	0.2797	2.55
S120 · near Wagholi	sample first	unusually green water	2026-10	12.11	3.3	-0.1983	-2.03	0.2213	2.81
S086 · city	sample first	unusually green water	2026-10	7.77	3.1	-0.2922	-0.85	0.2286	2.71
S109 · near Wadgaon-shinde	sample first	unusually green water	2026-10	4.95	0.0	-0.2361	-1.83	0.1419	3.17
S013 · Marunji	watch		2026-10	6.07	38.4	-0.2438	-0.37	0.0836	2.13
S033 · Pashan	watch		2026-10	40.51	38.0	-0.1085	1.67	0.1471	-1.22
S094 · Charholi Kh.	watch		2026-10	2.7	37.3	-0.1571	-0.75	0.2303	1.97
S098 · near Nirgudi	watch		2026-10	2.09	30.8				
S084 · near Borhadewadi	watch		2026-10	2.09	26.9	-0.1255	2.28	0.1931	-2.01
S146 · Manjari Kh.	watch		2026-10	14.28	18.9	-0.1858	-0.43	0.2125	1.53
S083 · city	watch		2026-10	13.2	11.9	-0.1308	0.14	0.1433	2.41
S103 · near Wagholi	watch		2026-10	17.62	11.6	-0.1678	-1.37	0.1684	1.84
S073 · city	watch		2026-10	16.82	11.2	-0.1198	-0.78	0.1265	1.65
S145 · Markal	watch		2026-10	15.49	11.2	-0.1371	-0.35	0.1861	1.71
S115 · near Wagholi	watch		2026-10	11.99	10.7	-0.1426	-0.77	0.1453	1.67
S099 · near Wagholi	watch		2026-10	12.96	8.4	-0.1453	0.18	0.1659	2.0
S014 · Lavale	watch		2026-10	2.57	7.8	-0.2399	1.14	0.1292	1.97
S068 · near Borhadewadi	watch		2026-10	13.76	7.3	-0.0306	0.62	0.1556	1.94
S090 · near Alandi Rural	watch		2026-10	6.84	6.5	-0.1466	-0.75	0.2063	1.87

Stretch	Status	Why	Month	Water area (ha)	Hyacinth %	Turbidity (NDTI)	Turbidity vs normal (z)	Chlorophyll (NDCI)	Chlorophyll vs norma
S050 · near Pimpale Gurav	watch		2026-10	5.75	6.3	-0.1546	0.25	0.1616	1.77
S072 · near Borhadewadi	watch		2026-10	5.23	6.2	-0.0861	1.13	0.1012	1.65
S002 · Lavale	watch		2026-10	6.92	5.2	-0.162	0.72	0.0948	2.2
S117 · Pimpalgaon Tarf Chakan	watch		2026-10	3.42	4.7	-0.0536	1.09	0.1657	1.88
S001 · Mulkhed	watch		2026-10	5.31	4.6	-0.1778	0.3	0.105	2.08
S140 · Markal	watch		2026-10	8.77	4.6	-0.1333	0.13	0.2132	1.86
S127 · Manjari Kh.	watch		2026-10	12.8	4.4	-0.1879	-0.55	0.2206	1.51
S100 · Dhanore	watch		2026-10	7.04	3.4	-0.1324	0.49	0.1969	1.59
S080 · city	watch		2026-10	7.81	3.1	-0.0761	2.17	0.0983	0.3
S058 · near Pimpale Gurav	watch		2026-10	12.15	3.0	-0.1102	1.17	0.1016	1.5
S101 · Dhanore	watch		2026-10	2.86	2.8	-0.1134	-0.21	0.1799	1.95
S141 · near Manjari Kh.	watch		2026-10	7.2	2.8	-0.2149	-1.51	0.2353	1.86
S138 · Manjari Kh.	watch		2026-10	15.17	2.7	-0.2048	-1.21	0.2269	1.56
S156 · Kadamwak Wasti	watch		2026-10	21.97	2.6	-0.2237	-1.6	0.2744	2.37
S064 · city	watch		2026-10	9.9	2.4	-0.1801	-0.22	0.1929	1.99
S157 · Apti	watch		2026-10	11.79	2.4	-0.1698	-0.44	0.1902	2.14
S107 · near Wagholi	watch		2026-10	5.23	2.3	-0.2123	-2.48	0.2332	2.36
S123 · near Wagholi	watch		2026-10	5.11	1.6	-0.1719	-0.01	0.1862	1.57
S110 · Solu	watch		2026-10	8.25	1.5	-0.1898	-1.14	0.2264	2.09
S016 · Bhukum	watch		2026-10	27.07	1.3	-0.1755	2.31	0.0962	1.91
S061 · near Pimpale Gurav	watch		2026-10	9.7	1.2	-0.1309	1.63	0.1177	1.73
S059 · near Pimpale Gurav	watch		2026-10	14.12	0.6	-0.3325	-2.28	0.3668	1.58
S088 · Kelgaon	watch		2026-10	6.8	0.6	-0.1542	-0.13	0.227	1.69
S063 · near Pimpale Gurav	watch		2026-10	8.37	0.0	-0.1341	1.71	0.1587	1.89

Stretch	Status	Why	Month	Water area (ha)	Hyacinth %	Turbidity (NDTI)	Turbidity vs normal (z)	Chlorophyll (NDCI)	Chlorophyll vs norma
S066 · near Pimpale Gurav	watch		2026-10	7.85	0.0	-0.1224	1.53	0.1519	1.57
S106 · near Wadgaon-shinde	watch		2026-10	1.53	0.0	-0.2425	-1.87	0.1335	1.6
S071 · near Gujar Nimbalkarwadi	normal		2026-10	10.5	26.4				
S132 · Bhawadi	normal		2026-10	2.45	26.2				
S030 · near Pimpale Saudagar	normal		2026-10	6.6	24.4				
S019 · near Hinjavadi	normal		2026-10	6.56	22.7				
S010 · Man	normal		2026-10	2.98	21.6				
S097 · near Charholi Kh.	normal		2026-10	5.19	20.2				
S029 · near Pimpale Saudagar	normal		2026-10	5.39	18.7				
S048 · near Borhadewadi	normal		2026-10	1.73	18.6	-0.1622	1.06	0.0701	-2.05
S154 · Loni-kand	normal		2026-10	5.55	18.1	-0.1458	0.71	0.0911	1.25
S006 · Man	normal		2026-10	5.87	17.8				
S028 · near Bhugaon	normal		2026-10	3.22	16.2				
S060 · near Pimpale Gurav	normal		2026-10	36.17	16.0	-0.2812	-0.32	0.3787	1.27
S131 · Wagholi	normal		2026-10	3.34	15.7	-0.1955	-1.87	-0.0008	-1.18
S045 · Pimpale Saudagar	normal		2026-10	2.82	14.3				
S026 · near Hinjavadi	normal		2026-10	3.42	14.1				
S136 · Tulapur	normal		2026-10	7.69	14.1				
S128 · Bhawadi	normal		2026-10	8.53	13.2	-0.1946	-0.53	0.0559	0.41
S126 · Wagholi	normal		2026-10	7.73	11.5	-0.2365	-0.4	0.0785	-0.97
S134 · Bhawadi	normal		2026-10	5.75	11.2				
S008 · Bhukum	normal		2026-10	3.26	11.1	-0.1892	0.49	0.0633	0.35
S017 · Man	normal		2026-10	3.3	11.0				
S036 · near Pashan	normal		2026-10	4.47	10.8				

Stretch	Status	Why	Month	Water area (ha)	Hyacinth %	Turbidity (NDTI)	Turbidity vs normal (z)	Chlorophyll (NDCI)	Chlorophyll vs norma
S108 · Nirgudi	normal		2026-10	5.83	10.3	-0.1507		0.2098	
S025 · Bhugaon	normal		2026-10	5.95	10.1	-0.1682	0.7	0.0315	-1.02
S051 · Pimpale Gurav	normal		2026-10	2.41	10.0				
S087 · city	normal		2026-10	14.52	10.0	-0.131	0.28	0.1597	1.31

Data sources

Dataset	Provider	Used for	Licence / citation
Sentinel-2 L2A surface reflectance (harmonized), 10-20 m	European Space Agency / Copernicus	River channel, hyacinth cover, turbidity and chlorophyll indices	Copernicus open licence.
Cloud Score+ for Sentinel-2	Google	Cloud and shadow masking	CC BY 4.0. Pasquarella et al. (2023), CVPR Workshops
ESA WorldCover 2021 v200	European Space Agency	Excludes trees and mangroves from floating-weed cover	CC BY 4.0.
Normalized Difference Turbidity Index (NDTI)	Lacaux et al. (2007)	Turbidity / suspended matter proxy	Method.
Normalized Difference Chlorophyll Index (NDCI)	Mishra & Mishra (2012)	Chlorophyll / algal growth proxy	Method.
NASA ARSET: Monitoring Water Quality of Inland Lakes using Remote Sensing (2023)	NASA Applied Remote Sensing Training	Method basis: Sentinel-2 spectral indices in Earth Engine for turbidity and chlorophyll, and calibration against field samples	Training material.
NASA ARSET: Hyperspectral Data for Land and Coastal Systems, Q&A; part 3 (2021)	NASA Applied Remote Sensing Training	Bank pixels excluded (mixed land-water); local calibration of indices; limits of land-oriented atmospheric correction over water	Training material.
NASA ARSET: Mapping and Monitoring Lakes and Reservoirs (2021); Remote Sensing of Freshwater Habitats (2019)	NASA Applied Remote Sensing Training	Background on lake and river monitoring with satellite data	Training material.
Palacios et al. (2015), atmospheric correction over coastal and inland waters	Remote Sensing of Environment	Known limitation of land-oriented atmospheric correction over water	Paper.

Satellite indicators are optical proxies. Use them to choose where and when to take water samples, and to check whether weed removal or new sewage plants changed the river.